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Vu Hoang Nguyen Phan

Curriculum Vitae

vuphan314

Vu Phan is a Computer Science Ph.D. student of Prof. Moshe Vardi at Rice University, expecting to graduate in 2022 Spring. He solves computational counting problems with dynamic programming and knowledge compilation. (Vu Phan is a citizen of Vietnam. He was issued an F1 visa and is currently staying in Texas.)

Education

2017–2022 **Doctor of Philosophy**, *Computer Science*, Rice University, GPA 3.61.

2017–2019 **Master of Science**, *Computer Science*, Rice University, GPA 3.60.

2014–2017 **Dual Bachelor of Science**, *Computer Science and Math*, Texas Tech University, GPA 4.00.

Service

2020–2021 **Peer Reviewer**, *International Conference on Artificial Intelligence and Statistics (AISTATS)*.

Bibliography

Vu Phan is the corresponding author of the following publications (authors sorted by surnames).

Thesis

[Pha19] Vu H. N. Phan. "Weighted Model Counting with Algebraic Decision Diagrams". MS thesis. Rice University, 2019. URL: scholarship.rice.edu/handle/1911/107761.

Conference Papers

[DPV20a] Jeffrey M. Dudek, Vu H. N. Phan, and Moshe Y. Vardi. "DPMC: Weighted Model Counting by Dynamic Programming on Project-Join Trees". In: *International Conference on Principles and Practice of Constraint Programming (CP)*. 2020. URL: arxiv.org/abs/2008.08748.

[DPV20b] Jeffrey M. Dudek, Vu H. N. Phan, and Moshe Y. Vardi. "ADDMC: Weighted Model Counting with Algebraic Decision Diagrams". In: *AAAI Conference on Artificial Intelligence (AAAI)*. 2020. URL: arxiv.org/abs/1907.05000.

Workshop Paper

[Pha18] Vu H. N. Phan. "Syntactic Conditions for Antichain Property in Consistency Restoring Prolog". In: *Workshop on Answer Set Programming and Other Computing Paradigms (ASPOCP)*. 2018. URL: arxiv.org/abs/1809.09319.

Software Projects

2019–2020 **DPMC**, *Dynamic Programming for Model Counting*, `github.com/vardigroup/DPMC`.
- Implemented a competitive weighted #SAT framework in [DPV20a] with Jeffrey Dudek.

Supervisor: Moshe Vardi (Rice University).

2019–2020 **ADDMC**, *Algebraic Decision Diagrams for Model Counting*, `github.com/vardigroup/ADDMC`.
- Implemented a competitive weighted #SAT solver in [DPV20b; Pha19].

Supervisor: Moshe Vardi (Rice University).

2016–2018 **LED**, *Language of Effective Definitions*, `vuphan314.github.io/LED`.
- Translated the literate programming language `LED` into `SequenceL` and `LaTeX`.

Supervisor: Nelson Rushton (Texas Tech University).

2015–2016 **L**, *Logic*, `github.com/iensen/LtoASPtranslator`.
- Translated the logic programming language `L` into `Answer Set Prolog` with Evgenii Balai.

Supervisor: Nelson Rushton (Texas Tech University).

Programming Languages

Procedural **Python, C++, Java**.

Functional **Scala, Scheme**.

Graduate Courses

2021 **Spring**, Reasoning about Software.

2020 **Fall**, Bioinformatics: Sequence Analysis (A-).

2020 **Spring**, Statistical Machine Learning (B+).

2019 **Fall**, Computer Systems Architecture (A+).

2019 **Spring**, Compiler Construction (A-), Multi-Core Computing (B).

2018 **Fall**, Artificial Intelligence (A-).

2018 **Spring**, Programming Languages (B).

2017 **Fall**, Logic in Computer Science (B+).

Teaching Assistantships

2020 **Spring**, Programming Languages.

2019 **Fall**, Logic in Computer Science.

2018 **Fall**, Reasoning about Algorithms.

2018 **Spring**, Object-Oriented Programming and Design.

Awards

2017–2021 **Ph.D. Tuition and Stipend**, Rice University.

2015–2017 **Academic Achievement Scholarship**, Texas Tech University.

2014–2016 **Proven Achiever Scholarship**, Texas Tech University.